# Hideaki Takahashi

ht2673@columbia.edu | Koukyosyumei | Hideaki Takahashi

## Summary


- Third-year Ph.D. student at Columbia University researching **system security**, **formal verification**, and **AI**.
- Multiple first-authored papers in top-tier conferences (**IEEE S&P**, **CCS**, **AAMAS**, **CVPR**).
- Discovered **100+ confirmed zero-day bugs and CVEs** in widely used Web3, AI Agent, and Rust projects.
- Skilled open-source developer of widely adopted tools (**1.5k+ GitHub stars**).
- Awarded **five medals in Kaggle**, the world's largest platform for machine learning competitions.


## Education

**Columbia University in the City of New York** — *New York, United States*

PH.D (COMPUTER SCIENCE) — *Sep. 2024 - Present*

- Supervised by Prof. Junfeng Yang
- Member of The Center for Digital Finance and Technologies

**The University of Tokyo** — *Tokyo, Japan*

BACHELOR OF ARTS AND SCIENCES (INFORMATICS) — *Apr. 2019 - Mar. 2024*

- Supervised by Prof. Alex Fukunaga., GPA: 3.85/4.0

## Papers

[1] Hideaki Takahashi, Suman Jana, and Junfeng Yang. Efficient branch-and-bound testing and verification of zkvms. In *ACM SIGSAC Conference on Computer and Communications Security*, 2026. **Peer-reviewed @ CCS'26** (CORE Rank: A*, Acceptance Rate: TBD).

[2] Hideaki Takahashi*, Jihwan Kim, Suman Jana, and Junfeng Yang. zkFuzz: Foundation and Framework for Effective Fuzzing of Zero-Knowledge Circuits . In *2026 IEEE Symposium on Security and Privacy (SP)*, pages 919–938, Los Alamitos, CA, USA, May 2026. IEEE Computer Society. **Peer-reviewed @ IEEE S&P'26** (CORE Rank: A*, Acceptance Rate: 13%).

[3] Hideaki Takahashi* and Alex Fukunaga. On the transit obfuscation problem. In *International Conference on Autonomous Agents and Multi-Agent Systems*, 2024. **Peer-reviewed @ AAMAS'24** (CORE Rank: A*, Acceptance Rate: 20.7%).

[4] Tianyuan Zou, Zixuan Gu, Yu He, Hideaki Takahashi, Yang Liu, Guangnan Ye, and Ya-Qin Zhang. VFLAIR: A research library and benchmark for vertical federated learning. In *The Twelfth International Conference on Learning Representations*, 2024. **Peer-reviewed @ ICLR'24** (CORE Rank: A*, Acceptance Rate: 31.1%).

[5] Hideaki Takahashi*, JingJing Liu, and Yang Liu. Breaching fedMD: Image recovery via paired-logits inversion attack. In *Conference on Computer Vision and Pattern Recognition*, 2023. **Peer-reviewed @ CVPR'23** (CORE Rank: A*, Acceptance Rate: 25.8%).

[6] Sally Junsong Wang, Jianan Yao, Kexin Pei, Hideaki Takahashi, and Junfeng Yang. Detecting buggy contracts via smart testing. *arXiv preprint arXiv:2409.04597*, 2024.

[7] Hideaki Takahashi*. Aijack: Security and privacy risk simulator for machine learning. *arXiv preprint arXiv:2312.17667*, 2023.

[8] Hideaki Takahashi*, JingJing Liu, and Yang Liu. Eliminating label leakage in tree-based vertical federated learning. *arXiv preprint arXiv:2307.10318*, 2023.

[9] Hideaki Takahashi*, Kohei Ichikawa, and Keichi Takahashi. Difficulty of detecting overstated dataset size in federated learning. Technical Report 10, 2021. `http://id.nii.ac.jp/1001/00214220/`.

## Talks

**UC Berkeley Security Seminar** *United States*
Invited Speaker *Mar. 11 2026*
- Title: zkFuzz: Foundation and Framework for Effective Fuzzing of Zero-Knowledge Circuits

**Zero Knowledge Frontier Japan: vol.2** *Japan*
Invited Speaker *Nov. 29 2025*
- Title: Typical Bugs and Detection Methods in Implementing Zero-Knowledge Proof Constraints

## Research Experience

**Columbia University** *New York, United States*
Ph.D Student *Sep. 2024 - Present*
- Conducted research on the software testing [2, 6] under the supervision of Prof. Junfeng Yang.

**Fukunaga Lab, The University of Tokyo** *Tokyo, Japan*
Undergraduate Student *Apr. 2023 - Mar. 2024*
- Conducted research on the transit obfuscation problem [3], a new task of privacy-preserving AI planning, under the supervision of Prof. Alex Fukunaga.

**Institute for AI Industry Research, Tsinghua University** *Beijing, China*
Federated Learning & Privacy Computing Interns *Jan. 2022 - Feb. 2023*
- Conducted research on federated learning and privacy computing [4, 5] under the supervision of Prof. Yang Liu and Prof. Jingjing Liu.

**Nara Institute of Science and Technology** *Nara, Japan*
Visiting Student *Aug. 2021 - Sep. 2021*
- Conducted research on the free-rider problem of federated learning [9] under the supervision of Prof. Kohei Ichikawa and Prof. Keichi Takahashi.

## Industry Experience

**Apple Inc.** *Yokohama, Japan*
Technical Internship: AIML/Software Engineer *Feb. 2024 - Jul. 2024*
- Worked on AIML/software engineering.

**UTokyo Economic Consulting Inc.** *Tokyo, Japan*
Research Assistant *Oct. 2020 - Mar. 2024*
- Worked on social implementations of econometrics and machine learning.

## Awards & Fundings

### Fundings

2025 **The Columbia-Ethereum Research Center for Blockchain Protocol Design**, PhD fellowship
2024 - 2026 **Funai Overseas Scholarship**, Granted 2 years of tuition and stipend.

### Competitions

2023 **45th / 616 teams (Silver Medal)**, Kaggle: Google - Fast or Slow? Predict AI Model Runtime
2021 **67th / 875 teams (Bronze Medal)**, Kaggle: Hungry Geese
2021 **52nd / 788 teams (Bronze Medal)**, Kaggle: Santa 2020 - The Candy Cane Contest
2020 **51st / 1138 teams (Silver Medal)**, Kaggle: Google Research Football with Manchester City F.C.
2020 **88th / 1390 teams (Bronze Medal)**, Kaggle: Cornell Birdcall Identification

## Software

**AIJack (`https://github.com/Koukyosyumei/AIJack`)**
Owner
- Security risk simulator for machine learning (400+ stars on GitHub, 10K+ downloads, referenced in 8+ papers)

**h5i (`https://github.com/h5i-dev/h5i`)**
Owner
- Axum-based Rust web framework designed for formal verification with Lean 4. (880+ stars on GitHub)

## Service

Reviewer

**Neruips 2026**
**IEEE Transactions on Network Science and Engineering**
**IEEE Transactions on Medical Imaging**
**IEEE Transactions on Dependable and Secure Computing**

## Skills

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|---|---|
| **Programming** | Rust, Lean 4, C, C++, CUDA, Python, Assembly, LLVM, Haskell, Solidity, Circom, R, Swift |
| **Languages** | English, Japanese |
| **DevOps** | AWS, Docker, GCP |